

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 3 -- Answer Key

Q1 Solution (30 points)

a) Communicating classes:

State i communicates with state j (written $i \leftrightarrow j$) if i can reach j AND j can reach i .

From 0: $0 \rightarrow 0$ (self), $0 \rightarrow 1$, $1 \rightarrow 0$. So $\{0, 1\}$ communicate.

From 0: 0 cannot reach 2 or 3 directly; $P(0,2)=0$, $P(0,3)=0$.

From 1: $1 \rightarrow 2$ but can 2 reach 1? $P(2,0)=P(2,1)=0$, so 2 CANNOT reach 1.

From 2: $2 \rightarrow 3$ and $3 \rightarrow 2$. So $\{2, 3\}$ communicate.

From 1: $1 \rightarrow 2$ (one step), but 2 cannot return to 1 or 0.

Communicating classes: $\{0, 1\}$ and $\{2, 3\}$.

b) Recurrent / Transient classification:

Class $\{2, 3\}$: once the chain enters states 2 or 3, all transitions stay within $\{2, 3\}$ ($P(2,0)=P(2,1)=P(3,0)=P(3,1)=0$). The chain NEVER leaves this class $\Rightarrow \{2, 3\}$ is RECURRENT (closed class).

Class $\{0, 1\}$: from state 1, $P(1,2)=0.5 > 0$, so there is a positive probability of jumping to $\{2,3\}$ and never returning. $\Rightarrow \{0, 1\}$ is TRANSIENT.

c) Stationary distribution of recurrent class $\{2,3\}$:

Solve $\pi(2) = 0.3\pi(2) + 0.7\pi(3)$ and $\pi(2) + \pi(3) = 1$:

$$0.7\pi(2) = 0.7\pi(3) \Rightarrow \pi(2) = \pi(3) = 1/2$$

For the transient class $\{0,1\}$: starting in $\{0,1\}$, the chain eventually leaves with probability 1 (since $\{0,1\}$ is transient). The chain transitions to $\{2,3\}$ and stays there forever.

Q2 Solution (35 points)

a) $E[X]$, $E[X^2]$, and long-run failure rate:

$$E[X] = 2 \cdot 0.4 + 5 \cdot 0.4 + 8 \cdot 0.2 = 0.8 + 2.0 + 1.6 = 4.4 \text{ days}$$

$$E[X^2] = 4 \cdot 0.4 + 25 \cdot 0.4 + 64 \cdot 0.2 = 1.6 + 10.0 + 12.8 = 24.4 \text{ days}^2$$

$$\text{Long-run failure rate} = 1/E[X] = 1/4.4 \sim 0.227 \text{ failures per day}$$

b) $P(N(10) \geq 5)$ using Poisson approximation:

$$m(10) \sim 10 / E[X] = 10 / 4.4 \sim 2.27$$

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$N(10)$ approx Poisson(2.27)

$$P(N(10) \geq 5) = 1 - P(N=0) - P(N=1) - P(N=2) - P(N=3) - P(N=4)$$

$$P(N=k) = e^{(-2.27)} * (2.27)^k / k!, \quad e^{(-2.27)} \sim 0.1033$$

$$P(N=0) \sim 0.1033, \quad P(N=1) \sim 0.2345, \quad P(N=2) \sim 0.2662$$

$$P(N=3) \sim 0.2011, \quad P(N=4) \sim 0.1141$$

$$P(N \geq 5) \sim 1 - 0.9192 \sim 0.0808 \quad (\text{about } 8.1\%)$$

c) Expected residual life $E[B]$ and Inspection Paradox:

$$E[B] = E[X^2] / (2E[X]) = 24.4 / (2 \cdot 4.4) = 24.4 / 8.8 \sim 2.77 \text{ days}$$

Compare $E[X]/2 = 4.4/2 = 2.2$ days.

Since $E[B] = 2.77 > 2.2 = E[X]/2$, the expected wait is longer than half the average gap.

Inspection Paradox: at a random time t , you are more likely to land inside a LONG inter-failure interval than a short one (because long intervals occupy more of the time axis). This biases the observed interval upward, so both the age $A(t)$ and the residual life $B(t)$ have mean $E[X^2]/(2E[X]) > E[X]/2$.

Q3 Solution (35 points)

a) $M(n) = S(n)^2 - n$ is a martingale:

$$\begin{aligned} E[M(n+1) \mid \mathcal{F}(n)] &= E[S(n+1)^2 - (n+1) \mid \mathcal{F}(n)] \\ &= E[(S(n) + X(n+1))^2 \mid \mathcal{F}(n)] - (n+1) \\ &= S(n)^2 + 2S(n)E[X(n+1)] + E[X(n+1)^2] - (n+1) \\ &= S(n)^2 + 0 + 1 - n - 1 \\ &= S(n)^2 - n = M(n) \quad (\text{OK}) \end{aligned}$$

b) $E[T]$ using OST on $M(n) = S(n)^2 - n$:

At stopping time T : $|S(T)| = N$, so $S(T)^2 = N^2$ always.

$$\begin{aligned} E[M(T)] &= E[M(0)] \Rightarrow E[S(T)^2 - T] = S(0)^2 - 0 = 0 \\ E[N^2 - T] &= 0 \Rightarrow E[T] = N^2 \end{aligned}$$

The expected time for a symmetric walk to first exit $(-N, N)$ is N^2 steps.

c) De Moivre's martingale -- verify $E[r^{X(n+1)}] = 1$:

$$r = q/p$$

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$$\begin{aligned}E[r^{X(n+1)}] &= p * r^{+1} + q * r^{-1} \\&= p * (q/p) + q * (p/q) \\&= q + p = 1 \quad (\text{OK})\end{aligned}$$

Since $E[r^{X(n+1)}] = 1$ and $X(n+1)$ is independent of $F(n)$:

$$\begin{aligned}E[M(n+1) \mid F(n)] &= E[r^{S(n+1)} \mid F(n)] \\&= E[r^{S(n)} * r^{X(n+1)} \mid F(n)] \\&= r^{S(n)} * E[r^{X(n+1)}] = r^{S(n)} * 1 = M(n) \quad (\text{OK})\end{aligned}$$